

Discussion of:

What Do 40 Trillion of Portfolio Holdings Say
about Monetary Policy Transmission?

by Chuk Fang and Kairong Xiao

Discussant:

Alan Moreira

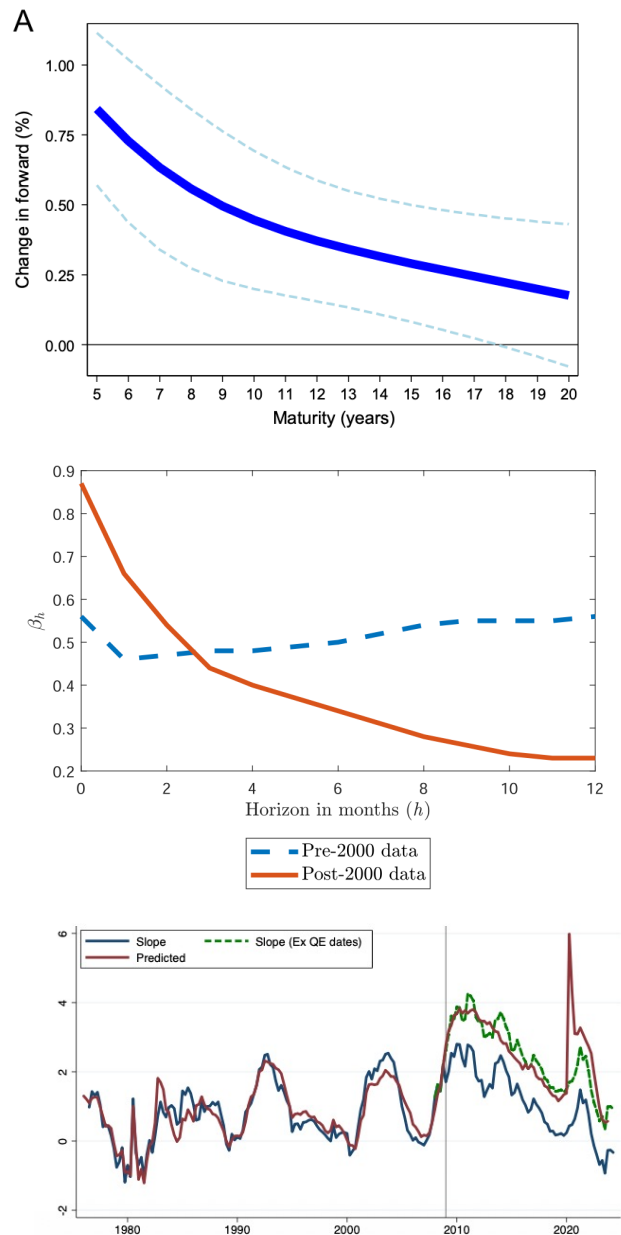
New York University and NBER



- Fed seems to have strong effects on all sort of markets
 - Stock Market – Bernanke Kutner (2005)
 - Long-term bonds – Hanson Stein (2015), Hilebrand (2023)
 - Exchange rates – Eichenbaum Evans (1995)
 - International stock markets- Miranda-Agrippino Rey (2020)
 - Deposits- Dreschler Savov Schnabl (2017)

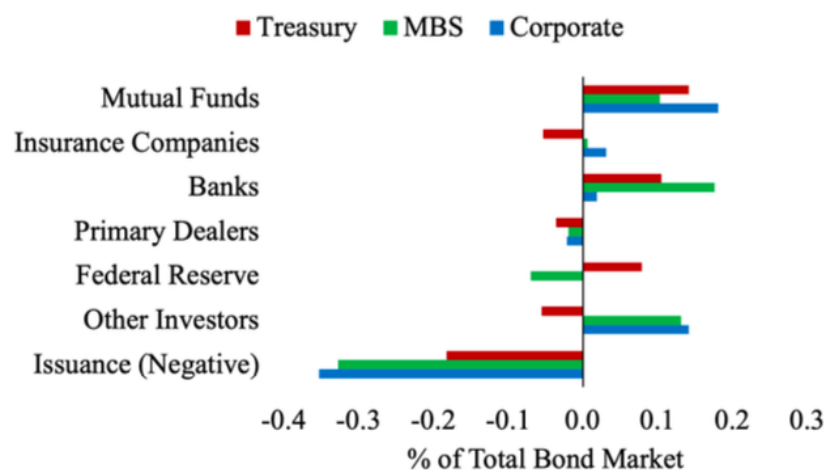
- Powerful effects extend also to QE/Asset purchases
 - Krishnamurty Vissing-Jorgensen (2011)
 - Haddad Moreira Muir (2021,2024,2025)

- Still unsettled why the fed is so powerful?
 - Information effects
 - Promises/ rules shocks
 - Agency frictions (reach for yield, ...)

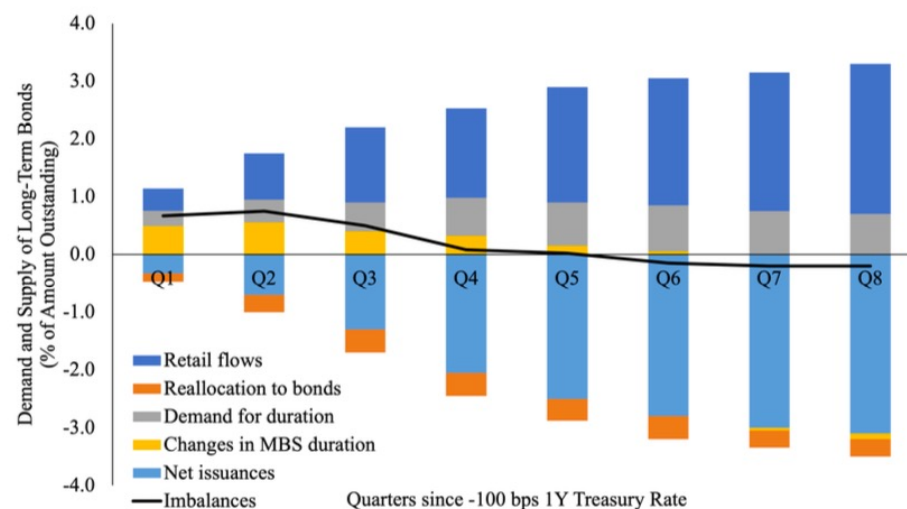


- Fact 1: the 10 year bond responds to movements in to “monetary-policy shock” (2 year around fomc) (Hanson Stein)
- Fact 2: These effect reverse within a year in the post-2000 sample (Hanson Lucca Wright)
- Fact 3: QE has very large effects
 - QE reduced 100-150bps of the term spread

- Issuances follow rate cuts, but slowly
- Interpret reversal as a result of this slow response of supply to the increase in demand for bonds



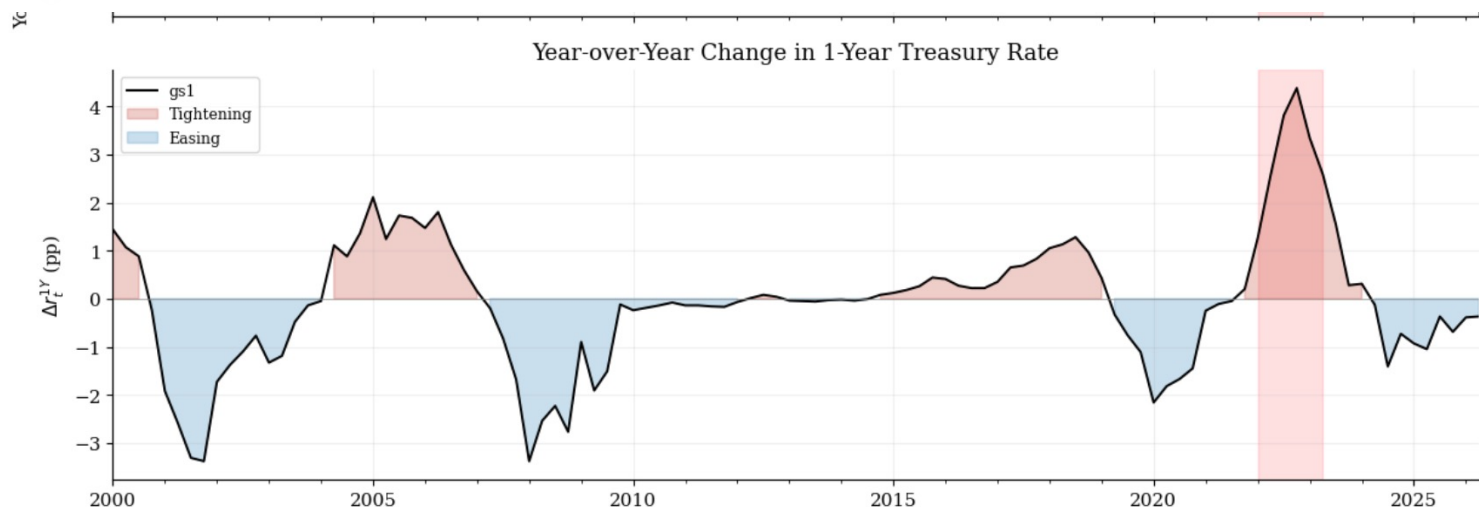
Reduced form



Structural

- In the end of the day, I don't think all the CUSIP level is helping at all
- This is an aggregate question and I am not sure why is helpful to think about very local substitution effects
- Likely getting those wrong harm the analysis, but at the very least they are distracting for this particular question

- How can we casually link the Fed, change in rates, and various quantity adjustments
 - Automatic stabilizers implies treasury issuance respond to weakening economic conditions –and so does the Fed by cutting rates
 - Corporations respond to increase uncertainty by borrowing more, just as the fed cuts rates
- I don't think controlling for VIX or past gdp is enough here
- Consider two years A and B: Same VIX/economic conditions in January
 - During year A rates dropped by 200bps
 - During year B , rates held steady
 - Which year has weaker economic conditions?



- The fact that issuance came later does not mean they were caused by the Fed
- The issuances can very much cause the fed to act
- Think about covid
 - Firms concerned with a sharp drop in revenues quickly tap all liquidity they can get
 - Credit lines, bond sales
 - Natural that they wanted more liquidity specially after seeing the market shutting down
- I don't see how the instruments discussed in the paper address these issues
- To be fair –the authors never claim that they do. I think they are more concerned with being able to estimate the demand curves and think of their instruments as largely about cross-sectional identification
- But even there I would appreciate a more robust discussion of why these instruments help

- How to think about delayed responses in issuance or arbitrage activity?
- How current market prices should respond?

FX: “Initially, a large gap emerges between demand and supply. ... This creates a significant initial imbalance that must be absorbed by a reduction in the term spread.”

- Currently the model has no way to connecting these dots. With ANY arbitrage activity an imbalance that is expected to close quickly will have only small effects
- How much a risk-premium should I get from half a year delay in the issuance response?
- Interesting ways forward:
 - Jansen Li Schmid augment the “DS” toolkit with a mean-variance arbitrageur
 - Hanson Lucca Wright write a dynamic model with slow moving capital → Limited arbitrage activity not sufficient to create the pattern

- In this sample the Fed-Treasury has crossed many Rubicons
 - QE in 2009
 - TARP
 - Corporate bond purchases in 2020
 - SVB/signature deposit bailouts
- Here is Chairman Bernanke on the eve of announcing QE1:

“Just crossing the Rubicon will have a significant announcement effect, because it will signal our willingness to do more if necessary in the future.”

- Haddad Moreira Muir show how a rule that leans against high term-premium
 - Creates significant crowding in—investors buying more as the Fed buys
 - Our paper attributes 2/3 of the 100 bps effect of QE on the term premium to the rule itself

- Imagine a world where one of the marginal investors cares about its portfolio variance. (s is the net supply)
$$E_t[r_{\tau,t+1}] + \frac{1}{2}\sigma_{\tau,t+1}^2 - r_t = \gamma \text{Cov}_t(r_{\tau,t+1}, s_t r_{p,t+1})$$

- Investors will demand a premium that depends on the supply

$$y_{\tau,t} - y_{\tau,t}^{s=0} = \gamma E_t \left[\sum_{j=0}^{\tau} s_{t+j} \text{Cov}_{t+j}(r_{\tau-j,t+j+1}, r_{p,t+j+1}) \right]$$

- In this world if you have clean net supply shocks—say QE changes the expected path of treasuries but nothing else, then event study identifies

$$- \sum_{j=0}^{\tau} \left(E_t[qe_{t+j}] E_t[\text{cov}_{t+j, qe}^{(\tau-j,p)}] - E_{t-1}[qe_{t+j}] E_{t-1}[\text{cov}_{t+j, qe}^{(\tau-j,p)}] \right)$$

- If instead this announcement also contains information about a new policy regime you get two extra terms

$$- \sum_{j=0}^{\tau} \left(E_t[qe_{t+j}] E_t[\text{cov}_{t+j, qe}^{(\tau-j,p)}] + \text{cov}_t(qe_{t+j}, \text{cov}_{t+j, qe}^{(\tau-j,p)}) + E_t \left[s_{t+j} \left(\text{cov}_{t+j}^{(\tau-j,p)} - \text{cov}_{t+j, qe}^{(\tau-j,p)} \right) \right] \right)$$

- You will have large yield effects even if there is no news in the expected path of policy

- Data on Asset Purchases announcements show a striking pattern that effects are strongly concentrated on earlier announcements
- Even the extremely large interventions during covid had tiny effects on yields

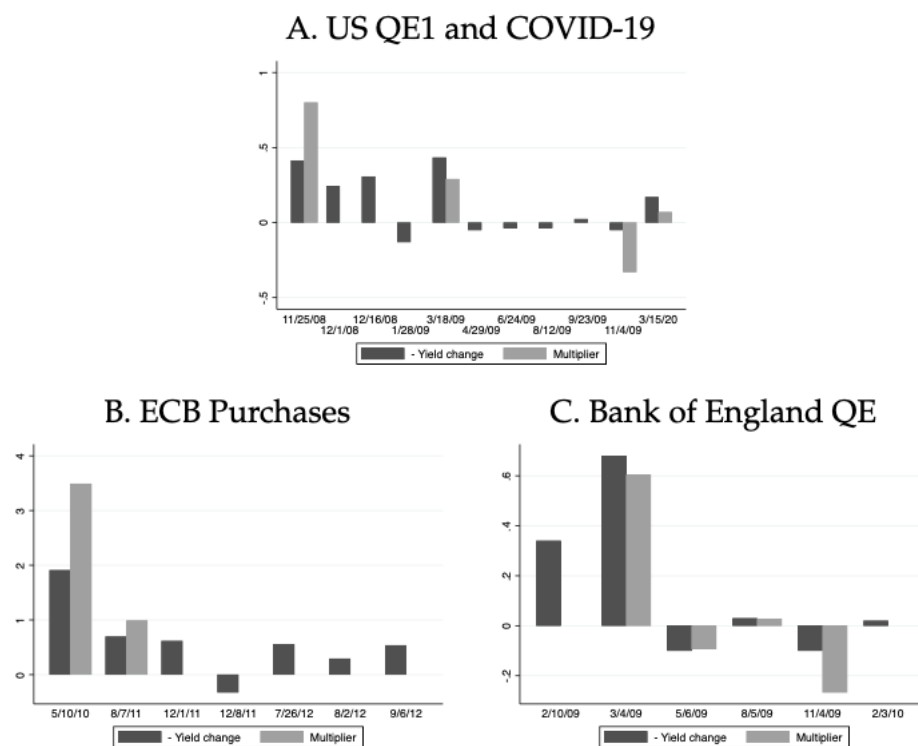


Figure 4: Weakening Announcement Effects of Asset Purchases

This figure is reproduced from Hatfield et al. (2022) and relates announcement effects to asset purchases.

- Even controlling for market expectations, early announcements are special
- Our view:
 - Early announcements are special because they communicate to the market a new policy regime (the two additional terms)
 - But if you mix both you end up with a parameter that is not obviously useful to any counterfactual
 - Cannot think about either the effect of policy shock or the systemic part of policy

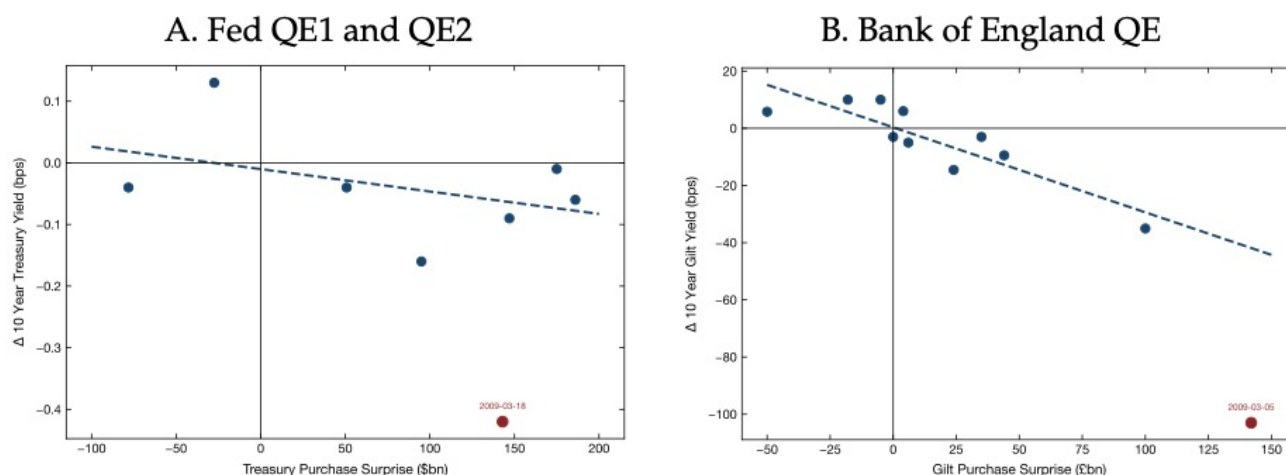
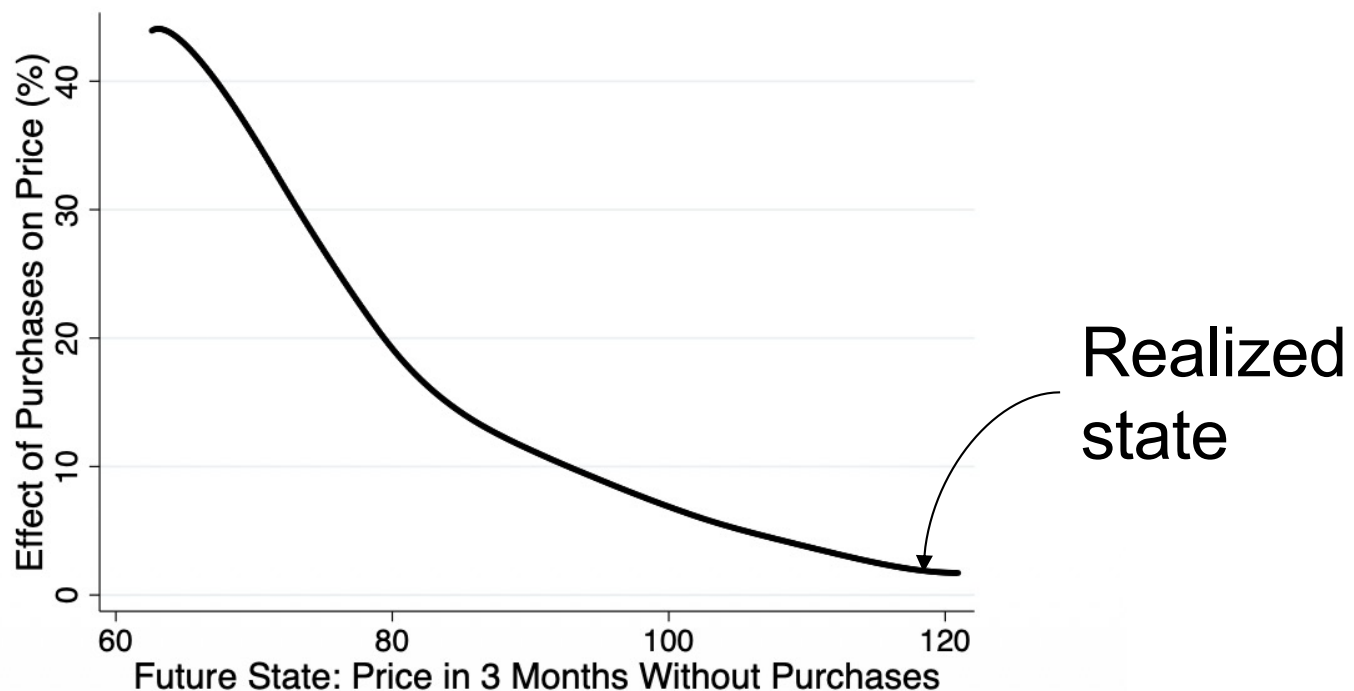


Figure 5: Survey Evidence on the Weakening Announcement Effects of Asset Purchases

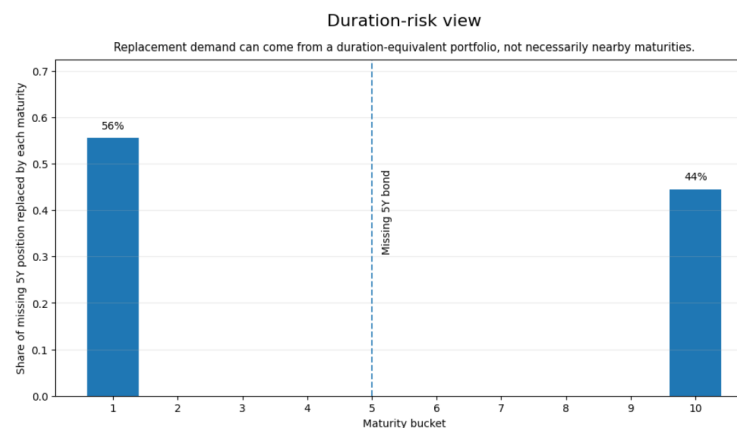
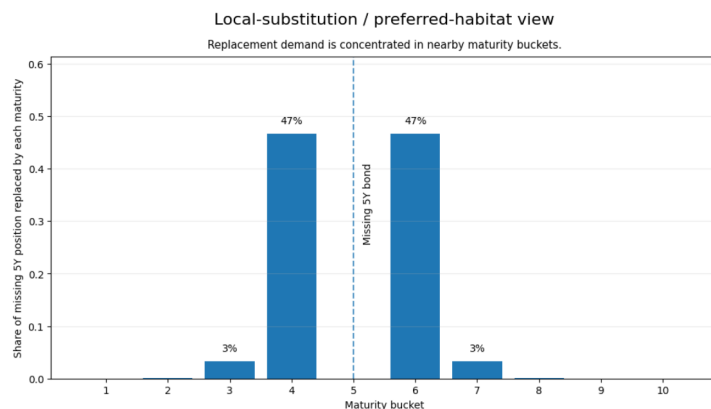
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- Using options we find that this exactly what the market believed



- Thinking hard about demand is obviously right
- The literature is doing a lot of progress in modeling substitution patterns
- This paper contributes by using random coefficients to allow heterogeneous substitution
 - But it is probably too “local”—it might be appropriate for some investors
 - But many of the stories we have in mind involve participants trading the factor exposures and not the characteristics
 - Pension funds manufacture a long-term IG bond by buying an IG bond and treasury future
 - Clearly duration is the right way to measure substitutability at the aggregate portfolio level for investors that do things like that
 - They might still have some local preferences between different IG bonds, but seems not central to think about the aggregate effects of QE



- Some criticisms of demand systems have been misguided in my view
 - We should be thinking about quantities and this rich data is amazing
 - We will have to make assumptions and they are never perfect
- But I wish there was more discussion if the “toolkit” fits the particular question
- Do we really need to model substitution across all bonds for this particular question?
- Discussion of identification should not be a side show
- Discussion of the variation being explored and why they allow you to think about interesting counterfactuals should be more central
- How can I use this particular elasticity that you estimated? Can the Fed use to design a new policy rule? Can it us to think about the effect of an one off policy shock?

Very ambitious paper

- Augment the KY toolkit
- Put together interesting data on the near universe of bond holdings
- First order question: Why Fed actions move markets so much?
- Results on mutual fund flows and issuances are super interesting and can speak to many theories of how the Fed impact markets